Linear Transformation

(To transform can from one form to another)

Let V and W be the 2 vectors space over a R.

A mapping $T: V \longrightarrow W$ is said to be linear transformation. If i) $T(x+y) = T(x) + T(y)$

ii) $T(\alpha x) = \alpha T(x)$

$\alpha \in R$

Problems :-

① Test whether the $T: R^3 \longrightarrow R^2$ defined by $T(x, y, z) = (x+y), (y+z)$ is linear or not.

Soln :- We need to prove i) $T(x+y) = T(x) + T(y)$

ii) $T(\alpha x) = \alpha T(x)$

where α is real number.

Let us consider $x = x_1, y_1, z_1$ and $y = x_2, y_2, z_2$

$$x+y = \underbrace{x_1+x_2}_{x}, \underbrace{y_1+y_2}_{y}, \underbrace{z_1+z_2}_{z}$$

$T(x+y, y+z)$

$$= T(x_1+x_2+y_1+y_2, y_1+y_2+z_1+z_2)$$

$$= T(x_1+y_1, y_1+z_1) + (x_2+y_2, y_2+z_2)$$

$$T(x+y) = T(x) + T(y) \longrightarrow ①$$

ii) $T(\alpha x) = T(\alpha x_1, \alpha y_1, \alpha z_1)$

$$= T[\alpha x_1 + \alpha y_1, \alpha y_1 + \alpha z_1]$$

$$= \alpha T[x_1+y_1, y_1+z_1]$$

$$T(\alpha x) = \alpha T(x) \longrightarrow ②$$

From ① and ②

∴ Hence T is a linear transformation.

② Verify the transformation $T : R^2 \longrightarrow R^2$ is defined by $T(x.y) = (x-y, x+y)$ is linear or not

Soln:- We need to prove i> $T(x+y) = T(x)+T(y)$

ii> $T(\alpha x) = \alpha T(x)$

Let us consider $x = x_1 . y_1$ and $y = x_2 . y_2$

$$x+y = \underbrace{x_1+x_2}_{x} . \underbrace{y_1+y_2}_{y}$$

$\to T(x-y . x+y)$

$$= T(x_1+x_2-y_1-y_2 . x_1+x_2+y_1+y_2)$$

$$= T(x_1-y_1 . x_1+y_1) + (x_2-y_2 . x_2+y_2)$$

$$T(x+y) = T(x) + T(y) \longrightarrow ①$$

ii> $T(\alpha x) = T(\alpha x_1 . \alpha y_1)$

$$= T[\alpha x_1 - \alpha y_1 , \alpha x_1 + \alpha y_1]$$

$$= \alpha T[x_1-y_1 . x_1+y_1]$$

$$T(\alpha x) = \alpha T(x) \longrightarrow ②$$

From ① & ②

∴ Hence. T is a linear transformation.

③ Show that $T(x.y) = (x+y . x-y , y)$ is a linear transformation or not.

Soln:- We need to prove i> $T(x+y) = T(x)+T(y)$

ii> $T(\alpha x) = \alpha T(x)$

Let us consider $x = x_1 . y_1$ and $y = x_2 . y_2$

$$x+y = x_1+x_2 . y_1+y_2$$

$$= T(x_1+x_2+y_1+y_2 , x_1-x_2-y_1-y_2 , y_1+y_2)$$

$$= T(x_1+y_1 , x_1-y_1 , y_1) + (x_2+y_2 . x_2-y_2 . y_2)$$

$T(x+y) = T(x) + T(y) \rightarrow ①$

ii) $T(\alpha x) = T(\alpha x_1, \alpha y_1)$

$= T[\alpha x_1 + \alpha y_1, \alpha x_1 - \alpha y_1, \alpha y_1)$

$= \alpha T(x) \rightarrow ②$

From ① & ②

Hence T is a linear transformation

Rank - Nullity theorem (without proof) :—

Let V and W be the vector space F and set T be the linear transformation from V to W i.e., $T : V \longrightarrow W$. If V is a finite dimension then $\boxed{\text{rank}(T) + \text{nullity of }(T) = \text{dimension of }(V).}$

Problems :—

① Let $T : R^3 \rightarrow R^3$ be the linear transformation defined by $T(x, y, z) = (x+y, x-y, 2x+z)$. Determine the range, null space and hence verify the rank-nullity theorem.

Soln :- consider the standard basis as a

$e_1 = (1.0.0)$, $e_2 = (0, 1. 0)$, $e_3 = (0. 0, 1)$.

Apply transformation on each

i.e., $T(e_1) = T(\underset{x}{1}.\underset{y}{0}.\underset{z}{0}) = (\underset{x+y}{1+0} \cdot \underset{x-y}{1-0}. \underset{2x+y}{2(1)+0})$

$T(1.0.0) = (1, 1, 2)$

$T(e_2) = T(\underset{x}{0}. \underset{y}{1}. \underset{z}{0}) = (0+1, 0-1. 2(0)+0)$

$T(0.1.0) = (1, -1, 0)$

$T(e_3) = T(\underset{x}{0}.\underset{y}{0}.\underset{z}{1}) = (0+0, 0-0, 2(0)+1)$

$T(0.0.1) = (0. 0. 1)$

$$A=\begin{bmatrix}1&1&2\\1&-1&0\\0&0&1\end{bmatrix} \quad R_2=R_2-R_1 \quad A=\begin{bmatrix}1&1&2\\0&-2&-2\\0&0&1\end{bmatrix}$$

$\therefore \text{rank}(T)=3$

$R_2 = \times^{ly}$ by $(-1/2)$

$$A=\begin{bmatrix}1&1&2\\0&1&1\\0&0&1\end{bmatrix}\begin{matrix}\to c_1\\\to c_2\\\to c_3\end{matrix}$$

To find the range $R(T)=\{c_1(1,1,2)+c_2(0,1,1)+c_3(0,0,1)\}$

range

$R(T)=\{(c_1, c_1+c_2, 2c_1+c_2+c_3)\}$

(to find the values of x.y.z)

To find the null space $n(T)$: $T(x.y.z)=0$

$T(x+y, x-y, 2x+z)=0$

$x+y=0 \qquad x-y=0 \qquad 2x+z=0$

$$\begin{array}{r}x+\cancel{y}=0\\x-\cancel{y}=0\\\hline 2x=0\\\boxed{x=0}\end{array} \qquad \begin{array}{r}x+y=0\\0+y=0\\\boxed{y=0}\end{array} \qquad \begin{array}{r}2(0)+z=0\\\boxed{z=0}\end{array}$$

$\therefore n(T)=0$

rank(T) + nullity of (T) = dimension of (V).

$3+0=3$

$\therefore$ Hence, verify the rank-nullity theorem.

② verify the rank-nullity theorem for the linear transformation $T: R^3 \to R^3$ defined by $T(x.y.z)=(x+2y-z, y+z, x+y-2z)$.

range. null space & verify rank-nullity.

last line becomes 0

rank$(T)=2$

$n(T)=1 \qquad x=3z, \; y=-z, \; z=k$

$\underline{sol^n}$:- consider the standard basis as a

$e_1 = (1.0.0)$, $e_2 = (0.1.0)$. $e_3 = (0.0.1)$

Apply transformation on each

i.e., $T(e_1) = T(\underset{x}{1}.\underset{y}{0}.\underset{z}{0}) = (1+2(0)-0,\ 0+0,\ 1+0-2(0))$

$T(1.0.0) = (1,0.1)$

$T(e_2) = T(\underset{x}{0}.\underset{y}{1}.\underset{z}{0}) = (0+2(1)-0,\ 1+0,\ 0+1-2(0))$

$T(0.1.0) = (2.1,1)$

$T(e_3) = T(\underset{x}{0}.\underset{y}{0}.\underset{z}{1}) = (0+2(0)-1,\ 0+1,\ 0+0-2(1))$

$T(0.0.1) = (-1,1,-2)$

$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix}$ $R_3 = R_3 + R_1$ $R_2 = R_2 - 2R_1$ $1-2(0)$ $1-2(1)$ $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$ $-1-1=$

$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{bmatrix}$ $R_3 = R_3 - R_1$ $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$

$R_3 = R_3 + R_2$ $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $\therefore$ rank$(T) = 2$

To find the range $R(T) = \{c_1(1.0.1) + c_2(2,1,1) + c_3(-1,1,-2)\}$

$R(T) = \{(c_1 + 2c_2 - c_3,\ c_2 + c_3,\ c_1 + c_2 - 2c_3)\}$

To find the null space $n(T)$: $T(x.y,z) = 0$

$T(x + 2y - z,\ y + z,\ x + y - 2z)$

$x + 2y - z = 0$, $y + z = 0$, $x + y - 2z = 0$

$3z + 2y - z = 0$ $\boxed{y = -z}$ $x - z - 2z = 0$

$3z - z = -2y$ $x - 3z = 0$

$2z = -2y$ $\boxed{x = 3z}$

$\boxed{z = \cdot z} = k$

$(3z, -z, z) = z(3, -1, 1)$.

$\therefore$ Nullity $(T) = 1$

→ By rank-nullity theorem

rank(T) + nullity(T) = dim(V)

$2 + 1 = 3$

Hence rank-nullity theorem verified

Matrix of linear transformation :-

* singular - [ ] ⇒ det = 0 → inverted of matrix exists.

In this topic we study how to associate a matrix to a linear transformation and also we study how to associate a linear transformation to a matrix.

Problems :-

① Find the matrix if $T: V_3(R) \longrightarrow V_2(R)$ is defined by $T(x, y, z) = (y - x, y - z)$

soln:- Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$

Apply transformation on each

$T(e_1) = T(\underset{x}{1}, \underset{y}{0}, \underset{z}{0}) = (-1, 0)$

$T(e_2) = T(\underset{x}{0}, \underset{y}{1}, \underset{z}{0}) = (1, 1)$

$T(e_3) = T(\underset{x}{0}, \underset{y}{0}, \underset{z}{1}) = (0, -1)$

The matrix of transformation is

$A = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}$ is $\begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$

(Invertable matrix)

② Find the matrix of the linear transformation $T: R^3 \longrightarrow R^2$ be defined by $T(x.y.z) = (2x+3y.\ y+2z)$.

Soln:- Let. $e_1 . e_2$ & $e_3 \in R^3$

$e_1 = (1.0.0),\quad e_2 = (0.1.0),\quad e_3 = (0.0.1)$

Apply linear transformation on each

$T(e_1) = T(1.0.0) = (2+0.\ 0+0) = (2.0)$

$T(e_2) = T(0.1.0) = (0+3.\ 0+0) = (3.0)$

$T(e_3) = T(0.0.1) = (0+0.\ 0+2) = (0.2)$

The matrix of transformation is

$$A = \begin{bmatrix} 2 & 0 \\ 3 & 1 \\ 0 & 2 \end{bmatrix} \text{ is } \begin{bmatrix} 2 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

③ $T(x,y) = (2x+y.\ x-2y)$

Soln:- Let. $e_1 = (1.0.0)$. $e_2 = (0.1.0)$. $e_3 = (0.0.1)$

Apply transformation on each

$T(e_1) = T(1.0.0) = T(2(1)+0.\ 1-2(0)$

$T(e_1) = t(2.1)$

$T(e_2) = T(0.1.0) = T(2(0)+1.\ 0-2(1)$

$T(e_2) = (1.\ -2)$

$T(e_3) = T(0.0.1) = T(2(0)+0.\ 0-2(0))$

$T(e_3) = (0.0)$

The matrix of transformation is

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -2 \\ 0 & 0 \end{bmatrix} \text{ is } \begin{bmatrix} 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix}$$

invertable matrix.

Singular & non-singular matrix :—

- det = 0 → singular [Kernel → span]
- det ≠ 0 → Non-singular – Invertable matrix exist

① Prove that $T: R^2 \to R^2$ is a singular matrix and find it's transformation if
$T(a, b) = (2a-4b, 3a-6b)$.

Given : $T: R^2 \to R^2$

We write matrix for the given transformation.

$$A = \begin{bmatrix} 2 & -4 \\ 3 & -6 \end{bmatrix} \qquad |A| = \begin{vmatrix} 2 & -4 \\ 3 & -6 \end{vmatrix}$$

$$= -12+12$$

$$|A| = 0$$

∴ The given transformation is singular because $\det(A) = 0$.

To find Kernel :- $T(a, b) = (0, 0)$

i.e $2a-4b = 0$

$3a-6b = 0$

After solving

$2a = 4b$ or $a = 2b$

$3a = 6b$ or $a = 2b$

Let, $b = K \Rightarrow \therefore a = 2K$

$\therefore Ker(F) = \{(2K, K)\}$

$Ker(F) = \{(a, b)\}$

$Ker(F) = Span(2, 1)$

(Module - 02)

1) Verify whether the linear transformation $T: R^3 \longrightarrow R^3$ defined by $T(x.y.z) = \{x+y+z, x+y-z, x-z\}$ is invertable or not.

Soln :- WKt, to check the transformation is invertable or not we need determinant of the given transformation.

$\begin{cases} * \text{ det} = 0 \rightarrow \text{Not invertable} \\ * \text{ det} \neq 0 \rightarrow \text{Invertable.} \end{cases}$

Let us consider basis e_1, e_2 and e_3 as a

$e_1 = (1.0.0)$, $e_2 = (0.1.0)$, $e_3 = (0.0.1)$

Apply transformation on each

$T(e_1) = T(\underset{x}{1}, \underset{y}{0}, \underset{z}{0}) = (1, 1, 1)$

$T(e_2) = T(0, 1, 0) = (1, 1, 0)$

$T(e_3) = T(0, 0, 1) = (1, -1, -1)$

The matrix of transformation is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

$$|A| = 1\begin{vmatrix} 1 & 0 \\ -1 & -1 \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$= 1(-1+0) - 1(-1-0) + 1(-1+1)$

$= \cancel{-1} \cancel{+1} -2$

$|A| = -2$

$|A| \neq 0$ $\therefore$ The transformation is invertible.

$\det(A) \neq 0$

2) Determine the linear transformation $T: R^2 \longrightarrow R^2$ such that $T(1, 1) = (0, 1)$ and $T(-1, 1) = (3, 2)$

Soln :- Given : $T(1, 1) = (0, 1)$ & $T(-1, 1) = (3, 2)$

$(x, y) = a(1, 1) + b(-1, 1) \longrightarrow *$

$(x, y) = (a-b, a+b)$

Equate the term

① $\leftarrow a - b = x$ and $a + b = y \rightarrow$ ②

$$\begin{array}{r} a - b = x \\ + \; a + b = y \\ \hline 2a = x + y \end{array} \qquad \begin{array}{r} a - b = x \\ - \; a + b = y \\ \hline -2b = x - y \end{array}$$

$a = x+y/2$ $\qquad$ $b = y-x/2$

Put a & b values in equation (*)

$$(x.y) = \frac{x+y}{2}(1.1) + \frac{y-x}{2}(-1.1)$$

Apply transformation on both sides

$$T(x.y) = \frac{x+y}{2}T(1.1) + \frac{y-x}{2}T(-1.1)$$

$$= \frac{x+y}{2}\underset{\text{given}}{(0.1)} + \frac{y-x}{2}\underset{\text{given}}{(3.2)}$$

$$= 0 + \frac{3}{2}(y-x),\ \frac{1}{2}(x+y) + \frac{y-x}{\cancel{2}} \times \cancel{2}$$

$$T(x.y) = \left\{\frac{3}{2}(y-x) + \frac{x+y}{2} + y-x\right\}$$

Question Bank

2) Test whether $T: R^3 \rightarrow R^3$ is linear transforma-tion defined by $T(x.y.z) = (2x+y.\ y-z.\ 2y+4z)$

Solⁿ:- We need to proove that i) $T(x+y) = T(x)+T(y)$

ii) $T(\alpha x) = \alpha T(x)$

Let us consider $x = x_1 . y_1 . z_1$ and $y = x_2 . y_2 . z_2$

i) $x+y = x_1+x_2 .\ y_1+y_2 .\ z_1+z_2$

$$= T(2x_1+2x_2+y_1+y_2 .\ y_1+y_2-z_1-z_2 .\ 2y_1+2y_2+4z_1+4z_2)$$

$$= \cancel{T(2x_1+y_1 .\ 2x_2+y_2) + (y_1-z_1 .\ y_2-z_2) +}$$

$$= T(2x_1+y_1 .\ y_1-z_1 .\ 2y_1+4z_1) + (2x_2+y_2 .\ y_2-z_2 .\ 2y_2+4z_2)$$

$T(x+y) = T(x)+T(y) \longrightarrow$ ①

~~ii) $T(\alpha x) = \alpha T(x)$~~

ii) $T(\alpha x) = T(\alpha x_1 .\ \alpha y_1 .\ \alpha z_1)$

$$= T(\alpha 2x_1+\alpha y_1 ,\ \alpha y_1-\alpha z_1 ,\ \alpha 2y_1+\alpha 4z_1)$$

$$= \alpha T(2x_1+y_1 .\ y_1-z_1 .\ 2y_1+4z_1)$$

$T(\alpha x) = \alpha T(x) \longrightarrow$ ②

~~From~~ ① and ②

Hence. T is a linear transformation.

3> $T(x.y.z) = (y-x.\ y-z)$

Sol^n:- We need to proove i) $T(x+y) = T(x)+T(y)$

ii) $T(\alpha x) = \alpha T(x)$

Let us consider $x = x_1 . y_1 . z_1$ and $y = x_2 . y_2 . z_2$

i) $x+y = x_1+x_2 .\ y_1+y_2 .\ z_1+z_2$

$= T(y_1+y_2-x_1-x_2 .\ y_1+y_2-z_1-z_2)$

$= T(y_1-x_1 , y_1-z_1) + (y_2-x_2 .\ y_2-z_2)$

$T(x+y) = T(x) + T(y) \longrightarrow ①$

ii) $T(\alpha x) = T(\alpha x_1 .\ \alpha y_1 .\ \alpha z_1)$

$= T(\alpha y_1 - \alpha x_1 ,\ \alpha y_1 - \alpha z_1)$

$= \alpha T(y_1 - x_1 ,\ y_1 - z_1)$

$T(\alpha x) = \alpha T(x) \rightarrow ②$

From ① and ②

Hence. T is a linear transformation.

4> $T(x.y.z) = (x-y+z,\ 2x-z .\ x+y-2z)$

Sol^n:- We need to proove i) $T(x+y) = T(x)+T(y)$

ii) $T(\alpha x) = \alpha T(x)$

Let us consider $x = x_1 . y_1 z_1$ and $y = x_2 . y_2 . z_2$

i) $x+y = x_1+x_2 .\ y_1+y_2 .\ z_1+z_2$

$= T[(x_1+x_2)-(y_1+y_2)+(z_1+z_2).\ 2(x_1+2x_2)-(z_1+z_2),$
$(x_1+x_2)+(y_1+y_2)-2(z_1+2z_2)]$

$= T(x_1-y_1+z_1 ,\ 2x_1-z_1 ,\ x_1+y_1-2z_1) +$
$(x_2+y_2+z_2 .\ 2x_2-z_2 .\ x_2+y_2-2z_2)$

$T(x+y) = T(x)+T(y) \longrightarrow ①$

ii) $T(\alpha x) = T(\alpha x_1, \alpha y_1, \alpha z_1)$

$= T(\alpha x_1 - \alpha y_1 + \alpha z_1,\ \alpha 2x_1 - \alpha z_1,\ \alpha x_1 + \alpha y_1 - \alpha 2z_1)$

$= \alpha T(x_1 - y_1 + z_1,\ 2x_1 - z_1,\ x_1 + y_1 - 2z_1)$

$T(\alpha x) = \alpha T(x) \longrightarrow (2)$

From (1) and (2)

Hence. $\underline{\underline{T \text{ is}}}$ a linear transformation.

5) Determine the matrix of the linear transformatio $T: V_2(R) \to V_3(R)$ defined by $T(x.y) = (x+y.\ x.\ 3x-y)$ with respect to $B_1 = \{(1.1), (-1.1)\}$ and $B_2 = \{(1.1.1), (1,-1.1), (0.0.1)\}$

$\underline{\text{soln}}$:- Given : $B_1 = \{\underset{}{\overset{u_1}{(1.1)}}, \overset{u_2}{(-1,1)}\}$

$B_2 = \{\underset{v_1}{(1.1.1)}, \underset{v_2}{(1,-1,1)}, \underset{v_3}{(0,0,1)}\}$

$T(u_1) = T(1.1) = T(1+1, 1, 3(1)-1) = (2,1,2)$

$T(u_2) = T(-1.1) = T(-1+1, -1, 3(-1)-1) = (0, -1, -4)$

Now we need to express the basic transformation as linear combn.

i.e., $a(1.1.1) + b(1.-1.1) + c(0.0.1) = 2.1.2$

$a+b+0 = 2 \Rightarrow a+b = 2 \rightarrow (1)$

$a-b+0 = 1 \Rightarrow a-b = 1 \rightarrow (2)$

$a+b+c = 2 \Rightarrow a+b+c = 2 \rightarrow (3)$

solve eqn (1) & (2)

$a+b = 2$

$a-b = 1$

$2a = 3$

$\boxed{a = 3/2}$

$\Rightarrow a+b = 2$

$\frac{3}{2} - b = 2$

$b = 2 - \frac{3}{2}$

$\boxed{b = \frac{1}{2}}$

$a+b+c = 2$

$\frac{3}{2} + \frac{1}{2} + c = 2$

$2 + c = 2$

$\boxed{c = 0}$

The matrix of linear transformation is

$$A \begin{bmatrix} 3/2 & -1/2 \\ 1/2 & 1/2 \\ 0 & 0 \end{bmatrix}$$

* $a(1.1.1) + b(1.-1.1) + c(0.0.1) = (0,-1.-4)$

$a+b = 0$

$a-b = -1$

$a+b+c = -4$

$a+b = 0$

(+) $a-b = -1$

$\boxed{a = -1/2}$

$a+b = 0$

$-\frac{1}{2} + b = 0$

$\boxed{b = 1/2}$

$a+b+c = 0$

$-1/2 + 1/2 + c = 0$

$\boxed{c = 0}$

9) Determine the matrix of the linear transformation $T: V_2(R) \to V_3(R)$ defined by $T(x,y) = (x+y, x, 3x-y)$ wrt standard basis.

Soln:- consider the standard basis

$e_1 = (1,0,0)$, $e_2 = (0,1,0)$ and $e_3 = (0,0,1)$

Apply transformation on each

$T(e_1) = T(\underset{x\ y\ z}{1,0,0}) = (1+0, 1, 3(1)-0) = (1, 1, 3)$

$T(e_2) = T(0,1,0) = (0+1, 0, 3(0)-1) = (1, 0, -1)$

$T(e_3) = T(0,0,1) = (0+0, 0, 3(0)-0) = (0,0,0)$

The matrix transformation is

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \text{ is } \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 3 & -1 & 0 \end{bmatrix}$$

Invertable matrix

18) Identify the rank nullity theorem of the linear transformation defined by $T(x,y,z) = (y-x, y-z)$

Soln:- consider the standard basis as a

$e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$

Apply transformation on each

$T(e_1) = T(1,0,0) = (0-1, 0-0) = (-1, 0)$

$T(e_2) = T(0,1,0) = (1-0, 1-0) = (1,1)$

$T(e_3) = T(0,0,1) = (0-0, 0-1) = (0,-1)$

$$\cancel{A = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}} \quad A = \begin{array}{ccc} c_1 & c_2 & c_3 \\ \end{array} \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$\therefore \rho(T) = 2$

To find the range $R(T) = \{c_1(-1,1,0) + c_2(0,1,-1)\}$

$R(T) = \{(-c_1,$

To find range $R(T) = \{c_1(-1,0) + c_2(1,1) + c_3(0,-1)\}$

$R(T) = \{(-c_1 + c_2 , c_2 - c_3)\}$

To find the null space $n(T)$: $T(x,y,z) = 0$

$T(y-x, y-z) = 0$

$y - x = 0$, $y - z = 0$

$\boxed{y = x}$. $\boxed{y = z}$

$\therefore$ Here $x = y = z$ $\Rightarrow$ The solⁿ is in the form of

$(K, K, K) = K(1, 1, 1)$

Hence, nullity $n(T) = 1$

By rank - nullity theorem

$\text{rank}(T) + \text{nullity}(T) = \dim(V)$

$2 + 1 = 3$ $\therefore$ LHS = RHS

Hence, the rank nullity theorem is verified.

15) verify the rank - nullity theorem $T: R^3 \to R^2$ be a linear transformation defined by

$T(x,y,z) = (x+y+2z, x-y-3z)$

Solⁿ:- consider the standard basis as a

$e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$

Apply transformation on each

$T(e_1) = T(1,0,0) = (1+0+2(0), 1-0-3(0)) = (1,1)$

$T(e_2) = T(\overset{x}{0}, \overset{y}{1}, \overset{z}{0}) = (0+1+2(0), 0-1-3(0) = (1,-1)$

$T(e_3) = T(0,0,1) = (0+0+2(1), 0-0-3(1) = (2,-3)$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & -3 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \end{bmatrix}$$

$\therefore \rho(T) = 2$

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To find the range $R(T) = \{C_i\}$

To find the null space $n(T)$: $T(x.y.z) = 0$

$T(x+y+2z), x-y-3z) = 0$

$x+y+2z = 0 \quad , \quad x-y-3z = 0$

$$\begin{array}{r} \cancel{x} + y + 2z = 0 \\ - \cancel{x} - y - 3z = 0 \\ (+) \\ \hline 2y + 5z = 0 \end{array}$$

$2y = -5z$

$\boxed{y = -\frac{5}{2}z}$

$x + \left(\frac{5}{2}\right)z - 3z = 0$

$x + 2.5z - 3z = 0$

$x - 0.5z = 0$

$\boxed{x = \frac{1}{2}z}$

$(x.y.z) = \left(\frac{1}{2}z, -\frac{5}{2}z, z\right)$

Put $z = 2$

$(x.y.z) = (1, -5, 2)$

Basis for the null space is $\{(1, -5, 2)\}$

Here. $n(T) = 1$

By rank-nullity theorem

$\text{rank}(T) + \text{nullity}(T) = \dim(V)$

$2 + 1 = 3$

Hence. the rank nullity theorem is verified.

13) If $T: V_3(R) \to V_4(R)$ defined by $T(1.0.0) = (0.1.0.2)$ $T(0.1.0) = (0,1,1.0)$ and $T(0.0.1) = (0,1,-1,4)$ identify the rank of the linear transformation.

Soln:- Given: $T(1.0.0) = (0.1.0.2)$

$T(0.1.0) = (0.1.1.0)$

$T(0.0.1) = (0.1.-1.4)$

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 4 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \quad A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$R_3 \to R_3 + R_2$

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here. there are 2 non-zero rows.

Hence. the rank of the linear transformation T is $\underline{\underline{2}}$.

Determine the matrix of linear transformation $T(x.y) = (-4x+2y.\ x,\ y-2x)$ with respect to the basis $B_1 = \{(1.1).(0.2)\}$ and $B_2 = \{(0.1.1).\ (1.0.1).\ (1.1.0)\}$

$\underline{\underline{Sol^n}}$:- $T(1.1) = (-2.1,-1) = x(0.1.1)+y(1.0.1)+z(1.1.0) \to (*)$

$y+z = -2 \to (1)$

$x+z = 1 \to (2)$

$x+y = -1 \to (3)$

Solve (1) & (2)

$y+z = -2$

$(-)\ x+z = 1$

$y-x = -3 \to (4)$

Solve (3) and (4)

$x+y = -1$

$-x+y = -3$

$2y = -4$

$\boxed{y=-2} \to$ Substitute in eq^n (3)

$x-2 = -1$

$\boxed{x=1}$

$T(0.2) = (4.0.2) = x(0.1.1)+y(1.0.1)+z(1.1.0) \to (*)$

$y+z = 4 \to (5)$

$x+z = 0 \to (6)$

$x+y = 2 \to (7)$

~~solve (5)~~

⑥ $\Rightarrow \boxed{x = -z}$ sub in ⑦

$-z + y = 2 \rightarrow$ ⑧

solve ⑤ and ⑧

$$\begin{array}{r} y + z = 4 \\ y - z = 2 \\ \hline 2y = 6 \end{array}$$

$$\boxed{y = 3}$$

By equation ⑤ $\rightarrow 3 + z = 4$

$$\boxed{z = 1}$$

The matrix of linear transformation is

$$\begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 1 \end{bmatrix}$$

verify the linear transformation $T: R^3 \rightarrow R^3$ defined by $T(x+y+z,\ x+y-z,\ x-z)$ is invertable or not.

$\underline{\text{soln}}$:- To verify the linear transformation is invertable we first form the matrix.

By writing the co-efficients of $x.y.z$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$|A| = 1\begin{vmatrix} 1 & -1 \\ 0 & -1 \end{vmatrix} - 1\begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$$

$$|A| = 1(-1+1) - 1(-1+1) + 1(0+0)$$

$|A| = 2 \neq 0$ $\therefore$ The given linear transformation

$\det(A) \neq 0$ is invertable.

$T(x.y) = (-4x+2y.\ x.\ y-2x)$

$B_1 = \{(1.1).\ (0.2)\}$, $B_2\{(0.1.1), (1.0.1).\ (1.1.0)\}$

<u>sol</u>^{n}:- $V = 1.1$ $\quad u = 0.2$

$T(V) = T(1.1) = T(-4+2,\ 1.\ 1-2) = (-2, 1, -1)$

$T(u) = T(0.2) = (4,\ 0.\ 2)$

Next we need to express the basis transforma^n.
as linear combination.

i.e., $a(0.1.1) + b(1.0.1) + c(1.1.0) = -2.1.-1$

$$0+b+c=-2 \Rightarrow b+c=-2 \rightarrow ①$$
$$a+0+c=1 \Rightarrow a+c=1 \rightarrow ②$$
$$a+b+0=-1 \Rightarrow a+b=-1 \rightarrow ③$$

solving eq^n ① & ②

$$b+\cancel{c} = -2$$
$$(-)a+\cancel{c} = 1$$
$$b-a = -3 \rightarrow ④$$

Solve eq^n ③ & ④

$$a+b=-1$$
$$(+)\ -a+b = -3$$
$$2b=-4$$
$$\boxed{b=-2}$$

$a+b=-1$, $a+c=1$

$a+(-2)=-1$ $\quad 1+c=1$

$\boxed{a=1}$ $\quad \boxed{c=0}$

$$A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

* $a(0.1.1) + b(1.0.1) + c(1.1.0) = (4.0.2)$

$$0+b+c=4 \Rightarrow b+c=4$$
$$a+0+c=1 \Rightarrow a+c=0$$
$$a+b+0=-1 \Rightarrow a+b=2$$

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \\ 0 & 1 \end{bmatrix}$$

$$b+\cancel{c}=4$$
$$a+\cancel{c}=0$$
$$b-a=4 \rightarrow ⑤$$

solve

$$\cancel{a}+b=2$$
$$\cancel{a}+b=4$$
$$2b=6$$
$$b=3$$

$a+b=2$

$a+3=2$

$\boxed{a=-1}$

$a+c=0$

$-1+c=0$

$\boxed{c=1}$

Determine the matrix of the linear transformation $T: V_2(R) \to V_3(R)$ defined by $T(x, y) = (x+y, x, 3x-y)$ wrt the basis $B_1 = \{(1, 1), (-1, 1)\}$ and $B_2 = \{(1, 1, 1), (1, -1, 1), (0, 0, 1)\}$

Soln :- $V = 1, 1$ $\quad u = -1, 1$

$T(V) = T(1, 1) = (2, 1, 2)$

$T(u) = T(-1, 1) = (0, -1, -4)$

Next we need to express the basis transform as linear combination i.e., $a(1.1.1)+b(1,-1,1)+c(0.0.1)=(2.1.2)$

$a+b+0=2 \Rightarrow a+b=2 \longrightarrow$ (1)

$a-b+0=1 \Rightarrow a-b=1 \longrightarrow$ (2)

$a+b+c=2 \longrightarrow$ (3)

solve eq^n (1) & (2)

$$\begin{array}{r} a+b=2 \\ +\ a-b=1 \\ \hline 2a=3 \end{array}$$

$a=3/2$

$\boxed{a=1.5}$

$a+b=2$

$1.5+b=2$

$b=2-1.5$

$\boxed{b=0.5}$

$a+b+c=2$

$1.5+0.5+c=2$

$2+c=2$

$\boxed{c=0}$

• $d(1,1,1)+e(-1,-1,1)+f(0,0,1)=(0,-1,-4)$

$d+e+0=0 \Rightarrow d+e=0 \rightarrow$ ④

$d+e+0=-1 \Rightarrow d-e=-1 \rightarrow$ ⑤

$d+e+f=-4 \Rightarrow d+e+f=-4 \rightarrow$ ⑥

Solve eqn ④ & ⑤

$$\begin{aligned} d+\cancel{e}&=0 \\ d-\cancel{e}&=-1 \\ \hline 2d&=-1 \\ d&=-1/2 \end{aligned}$$

$\boxed{d=-0.5}$

$d-e=-1$

$-0.5-e=-1$

$-e=-1+0.5$

$e=1/2$

$\boxed{e=0.5}$

$d+e+f=-4$

$-0.5+0.5+f=-4$

$\boxed{f=-4}$

The matrix of linear transformation is

$$A=\begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}=\begin{bmatrix} 1.5 & -0.5 \\ 0.5 & 0.5 \\ 0 & -4 \end{bmatrix}$$

Identify space range, rank and nullity of the linear transformation also verify the rank - nullity theorem defined by

$T(x_1, x_2)=(x_1+x_2, x_1)$

Soln:- Let us consider the basis

$e_1=(1,0), \quad e_2=(0,1)$

Apply the transformation on each basis

$T(e_1)=T(\underset{x_1\ x_2}{1,0})=(1,1)$

Nullity → Zero rows

$T(e_2)=T(0,1)=(1,0)$

$$A=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$R_2 \rightarrow R_2-R_1$

$$A=\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}\begin{matrix} \rightarrow x_1 \\ \rightarrow x_2 \end{matrix}$$

$R_1 \rightarrow R_2+R_1$

$\therefore \begin{cases} \rho(T)=2 \\ n(T)=0 \end{cases}$

$r(T) + n(T) = \dim(V)$

$2 + 0 = 2$

Hence the rank nullity theorem is verified.

To find out the range, $R(T)$

$$R(T) = \underset{\substack{\text{linear}\\ \text{comb}^n}}{L(S)} = \{x_1(1,1) + x_2(0,-1)\}$$

$$= \{x_1 + 0,\ x_1 - x_2\}$$

$$R(T) = \{(x_1, x_1 - x_2) \mid x_1, x_2 \in R\}$$